

AI-Assisted Acoustic Detection of Sperm Whales

A production-proven passive acoustic monitoring workflow

Overview. Passive acoustic monitoring (PAM) generates enormous volumes of underwater audio — far more than analysts can review by hand. We have developed and operationally deployed a machine-learning workflow that automatically detects and classifies marine mammal vocalizations in PAM recordings and returns explainable, analyst-reviewable results. The approach is proven in the field on Rice's whale (*Balaenoptera ricei*) at 96–98% accuracy and is directly extensible to sperm whale (*Physeter macrocephalus*) detection.

Why sperm whales fit this approach. Sperm whales are among the most acoustically active cetaceans, producing broadband impulsive echolocation clicks and stereotyped social "codas" almost continuously while foraging. These signals are distinctive and energetic — clicks render in a spectrogram as broadband vertical striations, codas as recognizable temporal patterns — making them well suited to image-based classification. Because the workflow is fully parameterized per species (sampling rate, frequency band, time–frequency resolution, color mapping), retargeting from Rice's whale's low-frequency calls to sperm whale's high-frequency click trains is a configuration-and-training exercise on labeled data, not a redesign.

The workflow — five stages:

1. **Audio ingest.** WAV/FLAC recordings from moored recorders, gliders, or drifting buoys are ingested and segmented into uniform time windows; long deployment files are chunked transparently for consistent analysis.
2. **Spectrogram generation.** Each segment is converted to a mel-scaled spectrogram tuned to the target species' acoustic band, with configurable noise reduction and color mapping, yielding standardized RGB images.
3. **GPU-accelerated classification.** Spectrograms are streamed in parallel — multi-threaded / multi-process I/O feeding batched GPU inference — through a deep convolutional neural network that scores each image as detection / no-detection with a confidence value. Throughput scales to hundreds of images per second on a single GPU.
4. **Explainable annotation.** For every detection the system produces a Grad-CAM heatmap overlay that highlights *where* in the time–frequency image the model identified the vocalization, with bounding boxes, alongside the raw spectrogram. Every machine decision is visually auditable by a human.
5. **Structured output & reporting.** Results are written to CSV (file, detection status, confidence, vocalization locations) and compiled into thumbnail PDF review reports — ready for QA/QC, archiving, or downstream occurrence and density analysis.

What partners get:

- **Scale** — automated triage of multi-month PAM deployments that would take analysts weeks to review manually.
- **Trust** — explainable AI (Grad-CAM) so detections are validated rather than black-boxed; essential for scientific and regulatory defensibility.
- **Transferability** — a species-agnostic pipeline with a documented production track record, proposed here for sperm whales and extensible to additional species (beaked whales, delphinids, and others) within the same framework.
- **Operational maturity** — an end-to-end deployment pipeline, not a research prototype.

Track record. The workflow is in operational use for Rice's whale detection in the Gulf of America, reaching 96–98% accuracy on field data through iterative retraining on real deployment recordings.